

Practical Cover Selection for Steganography

Zichi Wang, Xinpeng Zhang, and Zhenxing Qian

Abstract—This paper focus on a practical scenario of cover selection for steganography, in which a part of the available images of the sender have been processed to improve visual quality, e.g., contrast enhancement, image denoising. In this case, not only the embedding distortion (caused by steganography), but also the processing distortion (caused by processing) should be considered when selecting cover image. We propose a cover selection method to combine the two kinds of distortion together to measure the suitability for steganography of the available images. To calculate the processing distortion, a classifier is trained to distinguish the processed images with the original ones. The classifier is then used to measure the possibility of the existence of processing. High possibility means high processing distortion. In addition, the current distortion minimization framework designed for steganography is employed to calculate the embedding distortion. Finally, both kinds of distortion are combined to form the total distortion, and the image with the minimal total distortion is selected as cover. Using the selected image for embedding, high undetectability can be achieved.

Index Terms—Cover selection, steganography, distortion.

I. INTRODUCTION

Steganography is a technique to transmit secret data via public channels without causing suspicion [1,2]. Digital images are widely used as covers for steganography since they are widespread. To achieve steganography, the most popular approach slightly modifying the content of cover images to carry some secret data [3,4]. In modern steganography, the embedding distortion caused by modifications on a cover image is minimized to maximize the undetectability [5]. As the adversary, steganalysis aims to identify the existence of steganography by analyzing the images transmitted on public channels [6,7]. Modern steganalytic methods use supervised machine learning to investigate the models of the covers and the stegos. The features are extracted from a set of images to train a common steganalytic model, which is then used to distinguish suspicious images [8-10]. Then the ensemble classifier [8] is used to measure the property of feature sets [11]. Recently, deep learning based steganalysis also performs well [12-14], in which the feature extraction is joint with classifier.

When there are a number of images available, the sender

can select the most suitable image for steganography. The concept “most suitable” is relative. No matter how many cover images available, there are always some most suitable images compared with the others. Most of the existing cover selection methods empirically select cover images [15-19]. For example, the images with a larger number of changeable DCT coefficients [15] or with high visual quality are selected as covers [16]. In [17], the images with the most similar blocks to those of secret image are selected as covers. In [18] and [19], the images with the most texture and content complexity are selected preferentially. These empirical cover selection methods do not perform satisfactory security. In [20], available images are represented by the Gaussian mixture model, then the Fisher Information Matrix is calculated and mapped into a real value to evaluate the hiding ability of an image. The authors of [21] select the images with the minimal embedding distortion as covers. In [22], the MMD (maximum mean discrepancy) distance between the stego set and a clear arbitrary image set is kept not larger than a normal threshold during cover selection to resist pooled steganalysis.

In some scenarios, a part of the available images of the sender may have been processed using some operations which are widely employed to improve the visual quality of image, e.g., contrast enhancement, image denoising. In this case, as shown in Fig. 1, the statistical properties of the processed image are different from that of the original image. Specifically, the operations, which are used to improve the visual quality of image, enhance the image content and suppress texture details usually. However, the locations with texture details are advantageous to steganography since modifications can be well concealed. Without texture details, the modifications caused by steganography can be easily captured by steganalysis, since the stego image contains two kinds of distortion (processing distortion and embedding distortion). Therefore, the processed images are not suitable for steganography, and should be excluded from the candidates. However, existing cover-selection methods focus on the embedding distortion (caused by steganography) but ignore the processing distortion (caused by processing).



Fig. 1 Steganography with processed image.

In this paper, we propose a cover selection method which considers both the processing distortion and embedding distortion. The processing distortion is measured by the possibility of the existence of processing. Meanwhile, the embedding distortion is calculated using current distortion minimization framework. Both kinds of distortion are combined

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to form the total distortion. The image with the minimal total distortion is selected as cover. The contributions of this paper are listed as follows:

- (1) We demonstrate the flaw of existing cover-selection methods when the cover image has been processed to improve the visual quality, e.g., contrast enhancement, image denoising;
- (2) We propose a cover selection method which is secure in the scenario that a part of the available images have been processed. Experiments show that the undetectability of our method is higher than existing cover selection methods when examined by modern steganalytic tools.

II. EFFECT ON STEGANOGRAPHY CAUSED BY PROCESSING

This section gives the evidence that the undetectability of steganography will decreased when the cover image has been processed.

As discussed in Section I, the processed images are not suitable for steganography. To demonstrate this visibly, we conduct a group of experiments over image data set UCID [23] which contains 1338 color images sized 512×384. All the images are converted into grayscale as the original images. Then seven different image operations (Histogram equalization, Sharpen, Expansion, Erosion, Wiener filtering, Median filtering, and Average filtering) are executed on the images respectively. The obtained images are used as cover and embedded using the popular steganographic method HILL [24] with payload 0.5 bpp (bits per pixel) respectively to obtain the corresponding stego images. The undetectability of steganography is checked by the popular feature extraction method SRMQ1 [9] and ensemble classifier [11]. The undetectability comparison is shown in Table 1. Where P_E is the minimal total error with identical priors achieved on the testing sets,

$$P_E = \min_{P_{FA}} \left(\frac{P_{FA} + P_{MD}}{2} \right) \quad (1)$$

and P_{FA} and P_{MD} are the false alarm rate and missed detection rate respectively. Higher P_E means higher undetectability.

Table 1. Undetectability comparison of several kinds of images.

Kinds of cover images	P_E
Original	0.3534
Processed by Histogram equalization	0.0000
Processed by Sharpen (with “sobel” operator)	0.0065
Processed by Expansion (filter size: 3×3)	0.0000
Processed by Erosion (filter size: 3×3)	0.0000
Processed by Wiener filtering (filter size: 3×3)	0.0254
Processed by Median filtering (filter size: 3×3)	0.0000
Processed by Average filtering (filter size: 3×3)	0.0057

It can be seen from Table 1 that the undetectability of the

processed images is much lower than the clear ones. It means that the undetectability of steganography is decreased when the cover image has been processed. Furthermore, the values of P_E tested on the processed images are close to zero for all cases. That means the modification-trace caused by steganography can be discovered by modern steganalytic tools easily when the cover images are processed. Therefore, the processed images are not suitable for steganography, which should be excluded from the candidates. In next Section, a cover selection method is proposed to avoid selecting the processed images as cover and minimize the embedding distortion meanwhile.

III. PROPOSED COVER SELECTION METHOD

We propose a cover selection method for steganography, in which both the processing distortion and embedding distortion are considered. The architecture of our method is shown in Fig. 2. Given n available images $\{X_1, X_2, \dots, X_n\}$ for selection, the distortion of each image is calculated respectively, which contains the processing distortion and embedding distortion. Then the two kinds of distortion are combined to form the total distortion $\{D_1, D_2, \dots, D_n\}$. Finally, the image $X_{f(1)}$ which corresponding to the minimal total distortion $D_{f(1)}$ is selected as cover for steganography. The details are as follows.

A. Processing Distortion

The processing distortion is caused by processing, e.g., contrast enhancement, image denoising. It is demonstrated in Section II that the processed images are not suitable for steganography. Therefore, the processed images should be excluded from the candidates for embedding. To achieve this, we use the possibility of the existence of processing to measure processing distortion. High possibility means high processing distortion.

To calculate possibility of the existence of processing, a digital image processing classifier (called DIP classifier for abbreviation) is trained to distinguish the processed images with the original ones. Specifically, denote there are r original images $\{I_1, I_2, \dots, I_r\}$ possessed by the sender, and w kinds of operations are executed on $\{I_1, I_2, \dots, I_r\}$ respectively. Then $r \times w$ processed images $\{I'_1, I'_2, \dots, I'_{rw}\}$ can be produced.

After that, the features of $\{I_1, I_2, \dots, I_r\}$ and $\{I'_1, I'_2, \dots, I'_{rw}\}$ are extracted to train the DIP classifier. Where the feature extraction method we employed in this paper is SPAM [8] due to its low computational complexity. In SPAM, the differences between adjacent pixels are modeled using first-order and second-order Markov chains. Then the transition probability matrices of Markov chains are used as the final feature vector. The extracted features are feed on ensemble classifier [11] to form the DIP classifier. Ensemble classifier is equipped with a

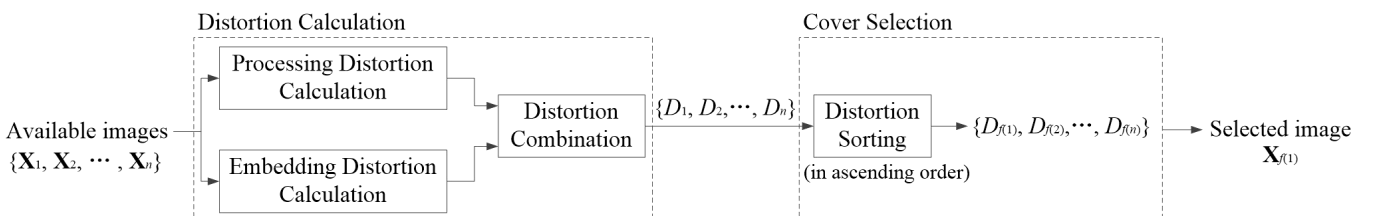


Fig. 2 Architecture of proposed method.

number of base learners (achieved by Fisher Linear Discriminant), and each base learner will make a binary decision (“original” or “processed”) on a given image. Each FLD learner is built on a randomly selected subspace of the feature space. Denote the number of base learners of the trained DIP classifier as S . Given an available image $\mathbf{X}_i$, $i \in \{1, 2, \dots, n\}$, assume there are L_i base learners made a “processed” decision, $L_i \in \{1, 2, \dots, n\}$.

Then the processing distortion D_i^P of $\mathbf{X}_i$ can be calculated as the rate of “processed” decisions as shown in Equation (2), since the rate stands for the possibility of the existence of processing in $\mathbf{X}_i$.

$$D_i^P = \frac{L_i}{S} \quad (2)$$

Note that $\{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n\}$ cannot be used for DIP classifier training, since a part of these images already have been processed and the indexes of processed images are unknown to the sender. Usually, n is much bigger than r , so there are more choices for cover selection among the available images. The sender used a small number of original images to train the DIP classifier which kept unchanged during cover selection. For the selection among the available images, the trained DIP classifier can help to identify the processed images. As a result, the undetectability of steganography can be guaranteed.

B. Embedding Distortion

For modern distortion minimization based steganographic methods, e.g., HILL [24], WOW [25], SUNIWARD [26], HDS [27], RBV [28], UT-GAN [29], etc, it is easy to calculate the embedding distortion as described in [22].

In modern steganography, a per-defined distortion function is used to assign embedding costs for cover elements to quantify the modification effect. Given an available image $\mathbf{X}_i$ contains t elements (pixels for an uncompressed image or DCT coefficients for a JPEG image) $\mathbf{X}_i = \{x_i(1), x_i(2), \dots, x_i(t)\}$, denote the embedding cost for +1 and -1 operation assigned for $x_i(j)$ as $\rho_i^+(j)$ and $\rho_i^-(j)$ respectively, $j \in \{1, 2, \dots, t\}$. For a payload limited sender, the theoretical minimal additive distortion d_i of the corresponding stego image with embedding capacity m (bits) [30] is,

$$d_i = \sum_{j=1}^t [p_i^+(j)\rho_i^+(j) + p_i^-(j)\rho_i^-(j)] \quad (3)$$

where

$$p_i^+(j) = \frac{e^{-\lambda\rho_i^+(j)}}{1 + e^{-\lambda\rho_i^+(j)} + e^{-\lambda\rho_i^-(j)}} \quad (4)$$

and

$$p_i^-(j) = \frac{e^{-\lambda\rho_i^-(j)}}{1 + e^{-\lambda\rho_i^+(j)} + e^{-\lambda\rho_i^-(j)}} \quad (5)$$

are the probabilities of modifying $x_i(j)$ by +1 and -1 respectively. The parameter λ ($\lambda > 0$) is used to make the ternary information entropy of modifying probability equal to the capacity m , as shown in Equation (6).

$$-\sum_{j=1}^t \{p_i^+(j)\log_2 p_i^+(j) + p_i^-(j)\log_2 p_i^-(j) + [1 - p_i^+(j) - p_i^-(j)]\log_2 [1 - p_i^+(j) - p_i^-(j)]\} = m \quad (6)$$

Therefore, the embedding distortion D_i^E of $\mathbf{X}_i$ can be measured by d_i calculated using Equation (3) directly. That means $D_i^E = d_i$.

C. Distortion Combination and Cover Selection

After the processing distortion and embedding distortion are obtained, the two kinds of distortion are combined to form the total distortion.

To combine the two kinds of distortion, multiplication is a good choice since the importance of these two kinds of distortion is equal. In addition, it can be known from Equation (2) that the range of processing distortion is between zero and one. When the value of processing distortion is zero, the processing distortion is nonexistent, but embedding distortion not. To avoid the total distortion becoming zero for the case that processing distortion is nonexistent. The total distortion D_i of $\mathbf{X}_i$ is defined as Equation (7) shown.

$$D_i = D_i^E (1 + D_i^P) \quad (7)$$

Finally, after sorting in ascending order, all the distortion $\{D_1, D_2, \dots, D_n\}$ are rearranged into $\{D_{f(1)}, D_{f(2)}, \dots, D_{f(n)}\}$. The image $\mathbf{X}_{f(1)}$ which corresponding to the minimal total distortion $D_{f(1)}$ is selected as cover for steganography. If it needs to select k images ($1 \leq k \leq n$), the k images corresponding to $\{D_{f(1)}, D_{f(2)}, \dots, D_{f(k)}\}$ are selected as cover. The highest distortion value acceptable for cover selection depends on the available images. We select the images with the minimal distortion values among the available images, since smaller distortion value means higher undetectability against steganalysis. The procedure of cover selection is summarized as ALG. 1.

ALG.1 Practical Cover selection.

Input: Available images: $\{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n\}$; Original images: $\{\mathbf{I}_1, \mathbf{I}_2, \dots, \mathbf{I}_r\}$; Number of needed cover images: k
Output: Selected images: $\{\mathbf{X}_{f(1)}, \mathbf{X}_{f(2)}, \dots, \mathbf{X}_{f(k)}\}$

- Train DIP classifier using $\{\mathbf{I}_1, \mathbf{I}_2, \dots, \mathbf{I}_r\}$, as described in subsection III.A;
 - Calculate processing distortion for all images in $\{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n\}$ using Equation (2) with the trained DIP classifier;
 - Calculate embedding distortion for all images in $\{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n\}$ using Equation (3);
 - Calculate total distortion for all images in $\{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n\}$ using Equation (7), then distortions $\{D_1, D_2, \dots, D_n\}$ are obtained;
 - Sort $\{D_1, D_2, \dots, D_n\}$ in ascending order, then distortions $\{D_{f(1)}, D_{f(2)}, \dots, D_{f(n)}\}$ are obtained and $\{\mathbf{X}_{f(1)}, \mathbf{X}_{f(2)}, \dots, \mathbf{X}_{f(k)}\}$ are the selected images;
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IV. EXPERIMENTAL RESULTS

A. Experiment Setup

To construct the available images of the sender, the 10000 grayscale images sized 512×512 in BOSSbase ver. 1.01 [31] are arbitrarily divided into ten subsets with the same size (1000 images in each subset). Then seven subsets are arbitrarily selected and processed by the seven operations: Histogram equalization, Sharpen, Expansion, Erosion, Wiener filtering, Median filtering, and Average filtering respectively, while the other three subsets are kept unchanged. All the obtained images are used as the available images ($n = 10000$).

for cover selection (seventy percent of the available images have been processed). Furthermore, the 1338 images in UCID [23] are used to train the DIP classifier ($r = 1338$). Specifically, all the 1338 images are processed by four operations: Histogram equalization, Sharpen, Expansion, and Erosion respectively ($w = 4$). Then features of the processed 5352 images and the original versions are extracted and used to train the DIP classifier. We use only four processing operations for the training of DIP classifier. This is reasonable since the sender does not know the kinds of operations used to process the available images in practical situation.

To verify the undetectability of our method, 500, 1000, 1500, 2000, and 2500 images are respectively selected from the available 10000 images using the proposed method as cover images ($k = 500, 1000, 1500, 2000, \text{ and } 2500$). For comparison, the same number of images are selected using the cover selection method in [21] and [22] respectively, and the same number of arbitrary images are also selected from the same available image set. Then the popular steganographic methods HILL is used for data embedding with payloads 0.1, 0.2, 0.3, 0.4, and 0.5 bpp respectively.

For steganalysis, the popular feature extraction methods SRM [9] and maxSRMd2 [10] are used to obtain the feature sets of cover and stego images. Then the ensemble classifier [11] is used to measure the property of feature sets, which is widely used for steganalysis. One-half of the cover and stego feature sets are used for training, while the remaining sets are used for testing. The criterion to evaluate the performance of feature sets is P_E as shown in Equation (1).

B. Undetectability Comparison

The comparisons of P_E for cover selection methods are shown in Fig. 3~Fig. 5 respectively. Where “HILL-Proposed”, and “SUNIWARD-Proposed” mean cover images are selected by our method and then embedded by HILL or SUNIWARD. “HILL-[21]”, “HILL-[22]”, “SUNIWARD-[21]”, “SUNIWARD-[22]” mean the cover images are selected by the method in [21] or [22], and embedded by HILL or SUNIWARD. While “HILL-Rand”, and “SUNIWARD-Rand” mean cover images are selected arbitrarily.

For different payloads, the comparison results are shown in Fig. 3 and Fig. 4. It can be seen that the undetectability of our scheme is much higher than other cover selection methods for all cases. This is reasonable since our method excluded the processed images as far as possible while other methods not. Actually, the undetectability of our method can be further increased by employing better feature extraction method for the training of DIP classifier. While the computational complexity [32] will also increase meanwhile. Our method is designed for modern distortion minimization based steganographic methods [24-29] which are the most popular steganographic methods. For other type of steganographic methods, our scheme also can be work by fixing the method used to calculate embedding costs.

For different number of selected images, as shown in Fig. 5, our method also performs the best undetectability. However, comparing with the method in [21], the advantage of our method is not remarkable when the number of selected images is large. It also can be predicted from Fig. 5 that the method in [21] is promising with a large number of selected images. In

fact, our method is designed to exclude the processed images and minimize the embedding distortion. When the number of selected images is large (such as 2500 images selected), there is not enough room to minimize the processing and embedding distortion meanwhile since only 3000 images are unprocessed among the available images. In fact, the performance of cover selection methods will be decreased with the increasing of the number of selected images, and equal to that for the case of arbitrary selection when all the available images are selected. Therefore, it is more significant to cover selection methods that perform well for small number of selected images.

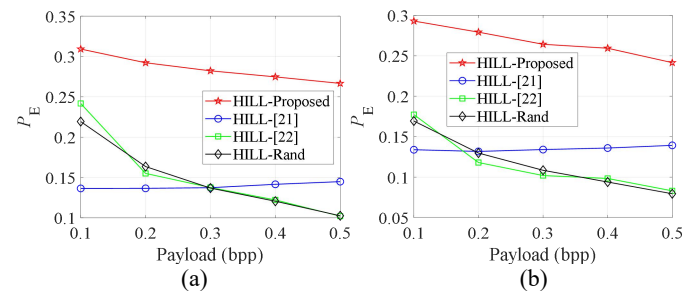


Fig. 3 Undetectability comparisons for cover selection methods on HILL with $k = 1000$ against (a) SRM and (b) maxSRMd2.

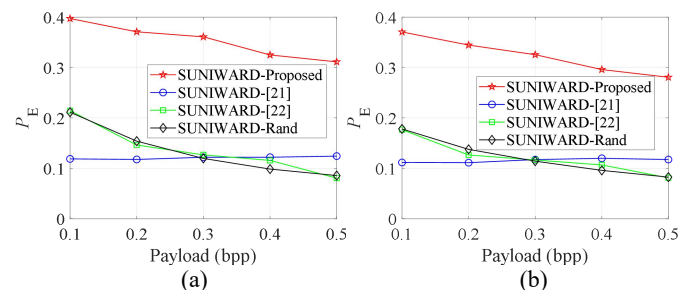


Fig. 4 Undetectability comparisons for cover selection methods on SUNIWARD with $k = 1000$ against (a) SRM and (b) maxSRMd2.

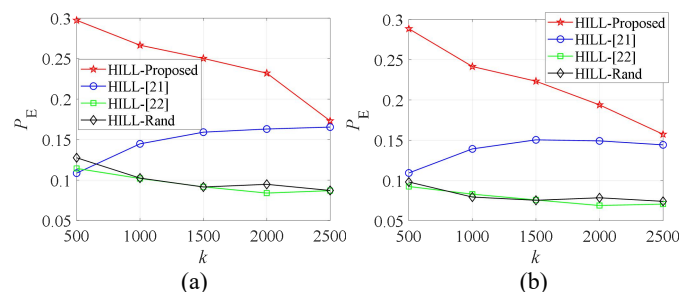


Fig. 5 Undetectability comparisons for cover selection methods on HILL with payload 0.5 bpp against (a) SRM and (b) maxSRMd2.

V. CONCLUSION

We investigate the disadvantage on steganography when the cover image has been processed, and then proposed a cover selection method which considers both the processing distortion and embedding distortion. Experimental results show that our method achieves the highest undetectability when examined by modern steganalytic tools. For further study, the processing distortion for steganography can be different for different kinds of processing operations. In addition, new steganographic methods can be developed to fit the properties of the process images. In this way, the process images also can be used for steganography.

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